

CONTROLLED FREQUENCY BREATHING REDUCES INSPIRATORY MUSCLE FATIGUE

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ABSTRACT

Burtch, AR, Ogle, BT, Sims, PA, Harms, CA, Symons, TB, Folz, RJ, and Zavorsky, GS. Controlled frequency breathing reduces inspiratory muscle fatigue. *J Strength Cond Res* 31(5): 1273–1281, 2017—Controlled frequency breathing (CFB) is a common swim training modality involving holding one's breath for approximately 7–10 strokes before taking another breath. We sought to examine the effects of CFB training on reducing respiratory muscle fatigue. Competitive college swimmers were randomly divided into either the CFB group that breathed every 7–10 strokes or a control group that breathed every 3–4 strokes. Twenty swimmers completed the study. The training intervention included 5–6 weeks (16 sessions) of 12 × 50-m repetitions with breathing 8–10 breaths per 50-m (control group) or 2–3 breaths per 50-m (CFB group). Inspiratory muscle fatigue was defined as the decrease in maximal inspiratory pressure (MIP) between rest and 46 seconds after a 200-yard freestyle swimming race (115 seconds [*SD* 7]). Aerobic capacity, pulmonary diffusing capacity, and running economy were also measured pre- and posttraining. Pooled results demonstrated a 12% decrease in MIP at 46 seconds post-race (−15 [*SD* 14] cm H₂O, effect size = −0.48, *p* < 0.01). After 4 weeks of training, only the CFB group prevented a decline in MIP values before to 46 seconds after race (−2 [13] cm H₂O, *p* > 0.05). However, swimming performance, aerobic capacity, pulmonary diffusing capacity, and running economy did not improve (*p* > 0.05) posttraining in either group. In conclusion, CFB training appears to prevent inspiratory muscle fatigue; yet, no difference was found in performance outcomes.

KEY WORDS running economy, pulmonary diffusing capacity, hypercapnia, hypoxia, athletes, respiratory muscle fatigue

INTRODUCTION

Swimming is often recognized as a sport in which the physiological demands may elicit pulmonary adaptations. Competitive swimmers exhibit larger static lung volumes and an enhanced diffusing capacity when compared with age- and height-matched control subjects (8). Commonly, swim coaches prescribe “hypoxic workouts,” which involve holding one's breath at total lung capacity (TLC) for approximately 7–12 strokes before taking another breath—a prolonged breath hold. This type of controlled frequency breathing (CFB) during a swimming workout promotes a hypercapnic rather than hypoxic effect (39). It is commonly accepted that, over time, this training may enhance fatigue resistance of the respiratory muscles (36), which may result in improved swimming times.

One study in particular has linked a performance benefit with CFB in recreational swimmers (i.e., novice triathletes). They found that CFB training over a 4-week intervention period improved 150-yard swim time by 8% (~13 seconds, effect size [*ES*] = 0.50) and reduced the oxygen cost of running (i.e., improved running economy) by 6% (a reduction of 15 ml·kg^{−1}·km^{−1}, *ES* = 0.57) in novice triathletes (23). In contrast, the group that took one breath every 2 strokes showed no improvement in either variable. This study, however, did not investigate the effects on inspiratory muscle fatigue, specifically.

The development of inspiratory muscle fatigue in swimming has been shown to occur after race-paced swimming over multiple distances and strokes (21,25–27). Inspiratory muscle training has been evaluated by multiple studies, and both pulmonary and performance benefits have been found after training (21,32,35,36). Multiple studies have investigated the effects of CFB (10,34,37,40); its effect on inspiratory muscle fatigue, however, has largely been uninvestigated.

The present study seeks to investigate the effects of CFB on inspiratory muscle fatigue in a competitive swim population. Nearly all the research on the development of inspiratory muscle fatigue exists in a “long-course meter” format (LCM, 50 m per length), whereas the National Collegiate Athletic Association (NCAA) race format is “short-course yards” (SCY, 25-yards per length). Therefore, it is

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important to describe the amount of fatigue that may develop over a similar race distance.

It was hypothesized that (a) a 200-yard (183 m) freestyle swimming race in competitive swimmers would elicit inspiratory muscle fatigue of the respiratory musculature as measured by maximal inspiratory pressure (MIP)/maximal expiratory pressure (MEP), (b) 4-week CFB training program would alleviate inspiratory muscle fatigue leading to faster swim times, and (c) a cross-training effect on terrestrial exercise (i.e., running) would occur after CFB training via reducing the oxygen cost of submaximal exercise.

METHODS

Experimental Approach to the Problem

To the best of the authors' knowledge, only one study has investigated the effects of CFB training on performance in an SCY format (23). In that study, the subjects were novice triathletes and the performance measure was a non-competition length of 150 yards (137 m). Currently, we seek to investigate the effects of 4 weeks CFB training on competitive swimmers at the NCAA level.

Measures of respiratory muscle fatigue were evaluated using MIP and MEP, taken at rest and immediately after a 200-yard (183 m) time trial. Respiratory muscle fatigue was quantified by a decline in MIP values after the time trial. After an intervention of 4 weeks CFB training, testing measures were repeated and data were analyzed for significant changes between the experimental and control groups. General spirometry, diffusing capacity of nitric oxide, and running economy were also evaluated pre- and postintervention to determine a possible physiological mechanism of improvement or any cross-training adaptations.

Subjects

Twenty-five competitive swimmers were recruited from the local University Swimming team. Members of this team were considered as competitive-level athletes having competed on a team placing in the top 10% of division I eligible men's and women's programs. Every subject was provided with an informed consent document explaining their responsibilities and risks by participating in this study. After questions were answered, the participants signed the document marking their entry into the study. This study was approved by the university's Institutional Review Board.

Training intervention occurred during the swimmers' pre-season to avoid any potential damages to the athletes' competition training schedule. Subject characteristics were as follows: The age ranged from 18 to 22 years old, with the mean and (*SD*) 20 (1) years old. Height was 177 (9) cm, weight was 77.6 (10.2) kg, body mass index was 23.1 (1.6) kg·m⁻², and body fat percentage was 16 (5) %. Body fat percentage was calculated as the average of two different equations (6,33). There was no statistical difference between groups in anthropometric data.

Procedures

Subjects were randomly assigned into 1 of 2 groups: experimental (CFB) or a control group (stroke matched). Subjects were required to perform a familiarization session along with 2 testing sessions. In the familiarization session, swimming performance, maximal inspiratory (MIP) and MEP, running economy, and aerobic capacity data were assessed. Baseline testing occurred 1 week later where the same data were collected (plus spirometry and diffusing capacity). Posttraining occurred at 4 weeks after baseline, including the same data collection.

Respiratory muscle strength was tested through a single point of maximal pressure development at the mouth (MIP and MEP). Testing procedure for MIP/MEP in the present study strictly followed American Thoracic Society (ATS)/European Respiratory Society (ERS) guidelines for the testing of volitional respiratory muscle strength published elsewhere (1). In accordance with these guidelines, a flanged mouthpiece was selected, no nose clips were used, and subjects were seated and were verbally encouraged to perform MIP and MEP efforts (Mueller and Valsalva) at or near residual volume and TLC, respectively.

A portable aneroid manometer (MicroRPM; Carefusion, Yorba Linda, CA, USA) linked to pulmonary management software (PUMA; Carefusion) was used to record the highest 1-second average measurement for each effort. Subjects were verbally encouraged to perform efforts lasting longer than 1.5 seconds in duration. After 3 efforts, the 2 highest efforts not differing more than 10 cm H₂O were recorded as the subject's maximal pressure. Thirty seconds rest was administered between efforts. The swimmers' resting MIP/MEP values during the baseline session were compared against predicted values (38).

After a generalized 1,000-yard swim warm-up, subjects were pretested for MIP and MEP. The athletes then performed a maximal 200-yard (183 m) freestyle time trial. No specific breathing pattern was administered. Additionally, breathing frequency was not counted during the time trial given no correlation exists with respect to the number of breaths and development of respiratory muscle fatigue, only swim intensity (28). Time was monitored on a S141 300 Lap Memory Stopwatch (Seiko, Tokyo, Japan), and all results from this session were both manually recorded and digitally stored as backups. MIP, followed by MEP, was tested at 3 separate points after the time trial to evaluate the depreciation in pressures and therefore the recovery rate of the respiratory muscles.

Spirometry, diffusing capacity, running economy, and aerobic capacity testing were done on a different day apart from the swimming time trials and MIP/MEP testing. Specifically, spirometry and diffusing capacity were completed at the same time of day for each subject between baseline and follow-up measures. For the measurement of oxygen and energy cost of running and aerobic capacity, the procedures set by Lavin et al. (2015) were followed using a calibrated Woodway ELG Treadmill (Woodway USA, Waukesha, WI, USA). A Parvo Medics TrueOne 2400 metabolic cart (Parvo Medics, Sandy,

TABLE 1. Participant performance characteristics at baseline.*

Variables	Control, SM (<i>n</i> = 12), mean (<i>SD</i>) (range)	Experimental, CFB (<i>n</i> = 13), mean (<i>SD</i>) (range)	Combined mean (<i>n</i> = 25), mean (<i>SD</i>) (range)	<i>p</i>
Swimming performance 200-yard time trial (s)	115 (7) (105–127)	114 (7) (105–129)	115 (7) (105–129)	0.61
Running treadmill data				
Peak heart rate (b·min ⁻¹)	191 (10) (179–210)	193 (9) (177–205)	192 (10) (177–210)	0.90
$\dot{V}O_{2\max}$ (L·min ⁻¹)	4.1 (0.9) (2.8–5.4)	4.2 (0.8) (3.0–5.1)	4.1 (0.8) (2.8–5.4)	0.83
$\dot{V}O_{2\max}$ (ml·kg ⁻¹ ·min ⁻¹)	52 (6) (39–63)	53 (5) (45–61)	53 (6) (39–63)	0.46
$VE_{\max}$ (L·min ⁻¹)	116 (26) (63–149)	114 (22) (81–149)	115 (2) (63–149)	0.81
$RER_{\max}$	1.12 (0.05) (1.01–1.18)	1.12 (0.05) (1.05–1.08)	1.12 (0.05) (1.01–1.18)	0.86
Running economy (mL·kg ⁻¹ ·km ⁻¹)	199 (10) (183–216)	203 (10) (183–217)	201 (10) (183–217)	0.32
Energy cost (kcal·kg ⁻¹ ·km ⁻¹)	0.99 (0.05) (0.90–1.08)	1.01 (0.05) (0.91–1.09)	1.0 (0.05) (0.90–1.09)	0.33

*SM = stroke matched; CFB = controlled frequency breathing; $\dot{V}O_{2\max}$ = maximal oxygen consumption; $VE_{\max}$ = maximal expired ventilation rate; $RER_{\max}$ = respiratory exchange ratio at maximal exercise.

UT, USA) was used for running economy and aerobic capacity assessments. Pulmonary function testing was conducted using a HypAir pulmonary function system (Medisoft, Dinant, Belgium) while seated in a standard office chair. Spirometry was measured according to ATS/ERS standardization of spirometry guidelines (30). Forced vital capacity, forced expiratory volume in 1 second, forced expiratory flow rate over the middle half of expiration (FEF_{25–75}), and peak expiratory flow rate were measured as part of the spirometry battery. The subjects' values were compared against reference equations (13). Pulmonary diffusing capacities for nitric oxide (DLNO) and carbon monoxide (DLCO) were also measured according to the methods described elsewhere (42), and subjects' DLNO and DLCO values were also compared against predictive equations (42). Pulmonary capillary blood volume (*V*_c) was determined based on the following: alveolar PO₂ (P_AO₂) = 100 mm Hg (42), the blood transfer conductance for NO (θNO) = 4.5 ml NO·min⁻¹·mm Hg⁻¹·ml⁻¹ of blood (4,7), blood transfer conductance for CO (θCO) = 0.584 ml CO·min⁻¹·mm Hg⁻¹·ml⁻¹ of blood where male hemoglobin concentration (Hb) = 14.6 g·dl⁻¹, and 0.537 ml CO·min⁻¹·mm Hg⁻¹·ml⁻¹ of blood where female (Hb) = 13.4 g·dl⁻¹. This was estimated using an equation to determine the blood transfer conductance equation for CO (θCO) (12): 1/θCO = (1.3 + 0.0041 P_AO₂) · (14.6/subject's Hb). Furthermore, the alveolar membrane diffusing capacity for carbon monoxide (DmCO) was calculated as the alveolar membrane diffusing capacity for nitric oxide (DmNO) divided by 1.97. Thus, DLNO < DmNO (4,5,41). The ratio of DLNO to DLCO was assumed to be an adequate surrogate for the DmCO to *V*_c ratio (16).

Subjects were instructed to attend 4 training sessions per week for 4 weeks. It should be noted that the timing of this study limits the athletes' potential to perform a time trial within 5–10% of their personal best. Each session lasted approximately 35 minutes; each subject underwent a standardized 1,000-m warm-up of easy mixed swimming. The training intervention occurred in a pool setup for LCM format to provide the athletes with a greater challenge. The workout consisted of 12 repetitions of a 50-m swim. The swimmers started each 50-m repetition at every minute for the first week, every 55 seconds for weeks 2 and 3, and progressing to 50 seconds per repetition by the fourth week. Only breaths taken while swimming each repetition were considered countable breaths during data collection. The CFB group was encouraged to limit their breathing to 2 breaths per lap resulting in approximately 24 breaths per workout. The control group was asked to breathe on a stroke-matched basis, breathing every 2–3 strokes accumulating to 10–12 breaths per lap. At the end of each workout, each subject reported their number of breaths taken per lap during the workout along with a rating of perceived exertion (RPE) from 6 (no exertion) to 20 (maximal exertion). Training sessions were supervised by at least one member of the swimming team's coaching staff.

Statistical Analyses

Testing procedures were performed at 3 time points, including familiarization, baseline, and posttraining. Therefore, the research design was a pre- and posttest design with control group in which a convenient sample of competitive

TABLE 2. Changes in exercise treadmill data and swim performance between baseline and follow-up.*†

	Mean change (SD) (95% CI)	Effect size	SRM	t-Statistic	p
Stroke matched (control group) (n = 11)					
$\dot{V}O_2\text{max}$ ($\text{L} \cdot \text{min}^{-1}$)	-0.1 (0.2) (-0.2 to 0.1)	-0.07	-0.34	-0.98	0.35
$\dot{V}O_2\text{max}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	-1.2 (3.1) (-3.2 to 0.9)	-0.24	-0.37	-1.23	0.25
Running economy ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{km}^{-1}$)	-3.3 (8.5) (-8.8 to 2.1)	-0.33	-0.39	-1.35	0.20
Energy cost ($\text{kcal} \cdot \text{kg}^{-1} \cdot \text{km}^{-1}$)	-0.02 (0.04) (-0.04 to 0.01)	-0.30	-0.38	-1.32	0.21
200-yard swimming performance time (s)	+1.8 (4.6) (-1.5 to 5.1)	0.28	0.38	1.22	0.26
Controlled frequency breathing group (n = 9)					
$\dot{V}O_2\text{max}$ ($\text{L} \cdot \text{min}^{-1}$)	+0.1 (0.3) (-0.1 to 0.3)	0.10	0.25	0.76	0.47
$\dot{V}O_2\text{max}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	+1.1 (3.2) (-1.4 to 3.5)	0.25	0.19	0.56	0.59
Running economy ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{km}^{-1}$)	+1.8 (9.5) (-5.5 to 9.0)	0.17	0.19	0.06	0.59
Energy cost ($\text{kcal} \cdot \text{kg}^{-1} \cdot \text{km}^{-1}$)	+0.01 (0.05) (-0.03 to 0.05)	0.16	0.19	0.58	0.58
200-yard swimming performance time (s)	-0.1 (1.5) (-1.2 to 1.1)	-0.01	-0.05	-0.16	0.88

*CI = confidence interval; SRM = standardized response mean.

†Mean (SD) (95% CI). The mean change is follow-up (posttraining) minus baseline.

college swimmers was used. Given the fact that the athletes had to be coached during the training phase, this study was not setup as blind or double blind. To compare groups at baseline for standard physical and anthropometric characteristics, independent *t*-tests were performed. For variables that were not normally distributed, a Mann-Whitney *U*-test was used to compare groups.

To compare swimming times and resting MIP and MEP between the familiarization session and the baseline session, paired *t*-tests were used. If any of the paired variables were

not normally distributed (as determined by a Shapiro-Wilk's test), a Wilcoxon signed rank test was used instead. A 1-way repeated measures analyses of variance were used to assess mean changes in MIP and MEP values at 4 different time points during the baseline session. These time points were after the warm-up but before the race, ~45, ~80, and ~110 seconds post-race. To compare the pre-race MIP and MEP values with the 3 post-race measurements, a Bonferroni correction was performed post hoc. To examine the differences in change in MIP values between pre- and posttraining within each group, paired *t*-tests were performed.

To distinguish between the inherent week-to-week variability of a 200-yard time trial, from small real physiological change, the coefficient of variation for the 2 initial swimming time trials was calculated for each subject and averaged ($[SD/\text{mean}] \cdot 100$). Reproducibility in swimming performance was calculated by obtaining the square root of the residual mean square error obtained from a repeated measures analysis of variance obtained from the 2 initial time trials that were performed over a 1-week period. Both groups were placed together in this analysis because neither group as of that point was under the influence of the training protocols. The square root of the residual mean square

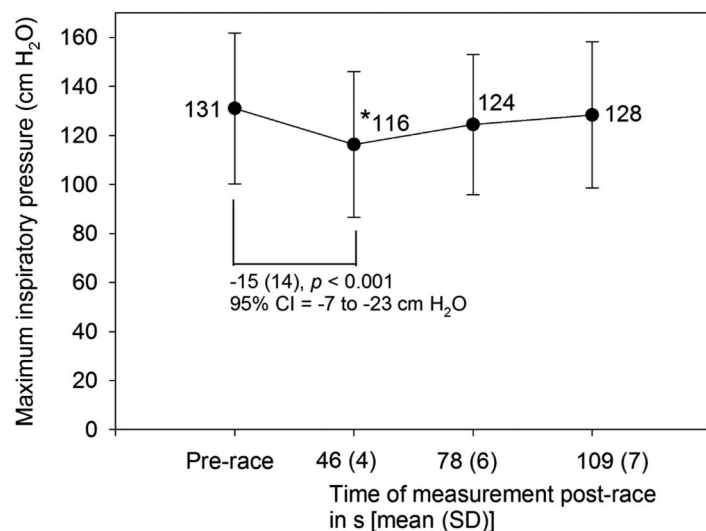
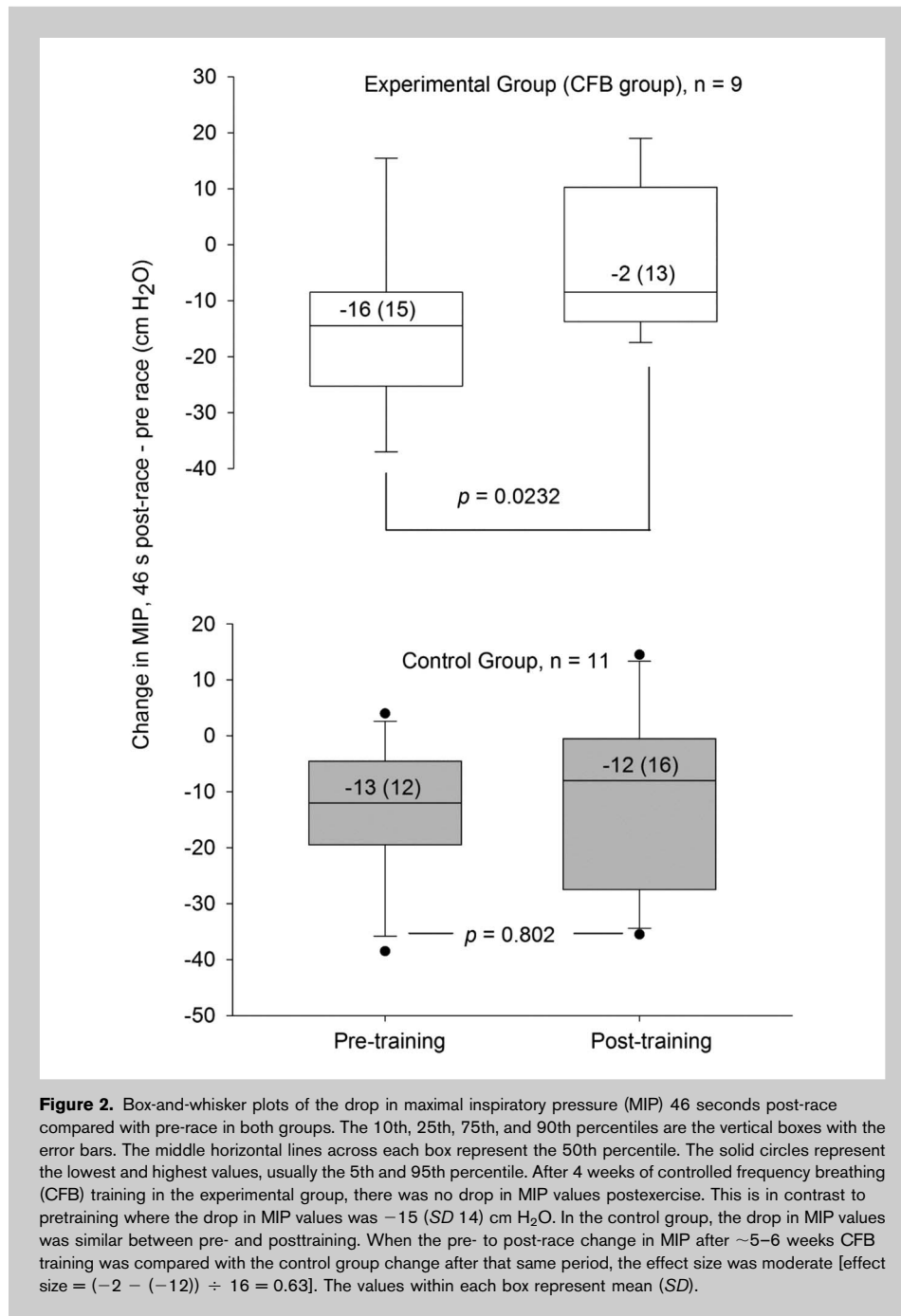


Figure 1. Evidence of global respiratory muscle fatigue from a 2-minute swim race. Approximately 46 seconds after a maximal swim effort of 200 yard (183 m), competitive swimmers demonstrate a significant ~12% drop in MIP values (effect size = -0.48). The MIP values recover by approximately 1.3 minutes postexercise.

*Statistically significant value from pre-race (n = 25, p < 0.001).



error obtained from the repeated measures analysis of variance was reported as the common week-to-week within-subject SD (SD_w) (3). Reproducibility was defined as $2.77 \cdot SD_w$ (Bland and Altman 1996). That is, the difference between the 200-yard swimming time trials on different weeks for the same subject is expected to be less than 2.77 times the SD_w for 95% of pairs of observations (3). Because the calculation of reproducibility may be considered too stringent, the smallest measurable change was reported as half of the reproducibility (15). Then, a Fisher's exact test was used to compare the

only 20 subjects completed the final testing session. There were 38 (SD 8) days between baseline testing and posttraining (follow-up) and 6 (1) days between familiarization and baseline. Each subject completed at least the minimum of 12 training sessions with a group average at 14 (2) sessions. For the CFB group, an average of 25 (3) breaths were taken per session for a RPE of 15 (1) out of 20. In the control group, 112 (9) breaths were taken per session and a RPE of 11 (1) out of 20. Both the number of breaths and the RPE were significantly different between groups ($p < 0.001$).

number of subjects in each group that improved time trial performance by more than the smallest measurable change. This was also done for aerobic capacity and running economy.

Indices of responsiveness to CFB training was calculated according to previous methods (22,31). Effect size was defined as the mean change of the variable between baseline and posttraining divided by the SD of the variable at baseline. An ES of 0.0–0.2 was considered trivial, 0.2–0.5 was small, 0.5–0.8 was moderate, and 0.8 and above was strong. The standardized response mean was calculated as the average change divided by the SD of the change. The t -statistic was defined as the mean change of the variable between baseline and posttraining divided by the standard error (which is the SD divided by the square root of the sample size).

The data were analyzed with the SPSS Statistical software package (SPSS Version 21.0; IBM SPSS Statistics, Inc., Chicago, IL, USA). Statistical significance was declared when $p \leq 0.05$ unless otherwise noted.

RESULTS

There were no differences between groups at baseline (Table 1). All subjects completed the familiarization and baseline sessions. However, because of travel, illness, and failure to comply with required number of training sessions,

Swimming Performance

There was a strong correlation ($r = 0.95$, $p < 0.001$) between the familiarization and baseline 200-yard swim trial times. The week-to-week variability swim time between the familiarization and baseline session was 1.4%. The SD_w was 1.5 seconds, the reproducibility was 4.3 seconds, and the smallest measurable change was 2.1 seconds. There was no statistical difference between mean swim time at either familiarization or baseline testing ($p = 0.262$). Four weeks of CFB training did not improve swim performance (Time $\times$ Group interaction, $p = 0.479$). The control group also did not significantly improve (Table 2). At baseline, the pooled mean swim time was 107% (3%) of the athlete's personal best 200-yard freestyle time trial; posttraining was 108% (5%) best. The number of subjects who improved by 2.1 seconds or more by the follow-up session was not different between groups (1 out of 9 for the CFB group vs. 3 out of 11 for the control group, Fisher's exact test, $p = 0.59$).

Respiratory Muscle Strength and Fatigue

Maximal inspiratory pressure increased by +7 (13) cm H₂O (95% confidence interval [CI], +2 to +13 cm H₂O, $p = 0.01$) between familiarization and baseline for an ~6% gain. Maximal expiratory pressure improved by +13 (20) cm H₂O (95% CI, +6 to +21 cm H₂O, $p < 0.01$) between familiarization and baseline for an ~10% gain. At baseline, MIP was 131 (31) cm H₂O, and MEP was 149 (27) cm H₂O, which were 133 and 117% of the predicted, respectively ($p \leq 0.05$ between measured and percent predicted).

Our baseline data demonstrated that an all-out 200-yard swim led to a 12% decrease in mean MIP scores at 46 (4) seconds post-race (Figure 1). The MIP value at 46 (4) seconds post-race was 116 (30) cm H₂O. However, after 1.3 minutes postexercise, mean MIP scores returned to near baseline values. After 4 weeks of CFB training, the drop in

mean MIP scores from pre- to 46 (4) seconds post-time trial was basically eliminated while the control group still experienced the same drop in mean MIP scores (Figure 2). The MEP scores did not change post-race at baseline or after the race posttraining only because MEPs were measured several minutes after the MIP tests.

Exercise Treadmill Test

The week-to-week variations in cardiopulmonary parameters from the running economy and maximal exercise treadmill test were the following: for running economy and the energy cost of running, the week-to-week variabilities were 2.4 and 2.5%, respectively, and 3.4 and 2.1% for aerobic capacity (liters per minute) and peak heart rate, respectively. The smallest measurable changes in aerobic capacity in liters per minute and milliliters per kilogram per minute, running economy, energy cost of running, and peak heart rate were 0.2 L·min⁻¹, 2.3 ml·kg⁻¹·min⁻¹, 6 ml·kg⁻¹·km⁻¹, 0.04 kcal·kg⁻¹·km⁻¹, and 5 b·min⁻¹, respectively. Four weeks of CFB training did not alter any of these parameters (Table 2). These parameters were also not altered in the control group (Table 2). Specifically, 5 out of 12 swimmers in the control group improved (reduced) their oxygen cost of running by at least 6 ml·kg⁻¹·km⁻¹ compared with their baseline values. This was no different from the CFB group (2 out of 9 swimmers) (Fisher's exact test, $p = 0.64$). There was no significant association between running economy and 200-yard swimming times ($n = 25$, $r = -0.26$, $p = 0.21$). However, ~65% of the variance in aerobic capacity (liters per minute) was accounted for by differences in swimming times.

Lung Function

At baseline, both male and female swimmers had larger lungs compared with predicted values. Forced vital capacity ranged from 112 to 163% of predicted (Table 3). At baseline,

TABLE 3. Baseline spirometry.*†

Variables	Female, n = 11	Female, % predicted	Male, n = 14	Male, % predicted	Combined, n = 25	Combined, % predicted
FEV ₁ (L)	4.3 (0.6) (3.8–5.6)	121 (10)† (104–140)	5.4 (0.4) (4.7–6.0)	110 (7)† (97–123)	4.9 (0.7) (3.8–6.0)	115 (10)† (97–140)
FVC (L)	5.4 (0.6) (4.7–6.5)	131 (14)† (112–163)	7.18 (0.48) (6.55–8.18)	121 (8)† (112–139)	6.4 (1.1) (4.7–8.2)	125 (12)† (112–163)
FEV ₁ /FVC	0.80 (0.06) (0.69–0.87)	93 (7)† (80–101)	0.76 (0.05) (0.69–0.86)	90 (5)† (82–102)	0.78 (0.06) (0.69–0.87)	91 (6)† (80–102)
PEF (L·s ⁻¹)	7.9 (1.2) (5.7–9.0)	110 (16)† (77–131)	10.1 (0.9) (8.2–11.4)	96 (10) (73–113)	9.1 (1.5) (5.7–11.4)	102 (15) (73–131)
FEF _{25–75} (L·s ⁻¹)	4.7 (0.9) (3.7–6.6)	121 (20)† (94–159)	5.6 (0.7) (4.5–7.1)	109 (14)† (90–139)	5.2 (0.9) (3.7–7.1)	114 (18)† (90–159)

*FEV₁ = forced expiratory volume in 1 second; FVC = forced vital capacity; PEF = peak expiratory flow; FEF_{25–75} = forced expiratory flow rate over the middle half of expiration.
†Mean (SD) (range) statistically significant within each sex.

TABLE 4. Changes in pulmonary diffusing capacity and components between baseline and follow-up.*†

	Mean change (SD) (95% CI)	Effect size	SRM	t-Statistic	p
Stroke matched (control group) (n = 9)					
DLCO (ml·min ⁻¹ ·mm Hg ⁻¹)	-1.2 (3.7) (-4.0 to 1.6)	-0.13	-0.32	-1.00	0.35
DLNO (ml·min ⁻¹ ·mm Hg ⁻¹)	+2 (16) (-10 to 15)	+0.05	+0.13	+0.46	0.66
Vc (ml)	+5 (10) (-2 to 12)	+0.30	+0.5	+1.61	0.15
DmCO (ml·min ⁻¹ ·mm Hg ⁻¹)	+17 (28) (-4 to 39)	+0.33	+0.61	+1.87	0.10
DLNO/DLCO ratio	+0.19 (0.31) (-0.05 to 0.43)	+0.54	+0.61	+1.86	0.10
Alveolar volume (L)	-0.1 (0.5) (-0.5 to 0.3)	-0.08	-0.20	-0.58	0.58
Controlled frequency breathing group (n = 9)					
DLCO (ml·min ⁻¹ ·mm Hg ⁻¹)	+1.5 (4.4) (-1.9 to 4.9)	+0.17	+0.34	+1.01	0.34
DLNO (ml·min ⁻¹ ·mm Hg ⁻¹)	+9 (25) (-10 to 28)	+0.20	+0.36	+1.05	0.32
Vc (ml)	-3 (10) (-10 to 5)	-0.12	+0.3	-0.79	0.45
DmCO (ml·min ⁻¹ ·mm Hg ⁻¹)	+13 (37) (-16 to 42)	+0.23	+0.61	+1.04	0.33
DLNO/DLCO ratio	+0.02 (0.28) (-0.20 to 0.23)	+0.09	+0.66	+0.18	0.86
Alveolar volume (L)	-0.1 (0.40) (-0.4 to 0.2)	-0.06	-0.38	-0.50	0.63

*CI = confidence interval; SRM = standardized response mean; DLCO = pulmonary diffusing capacity for carbon monoxide; DLNO = pulmonary diffusing capacity for nitric oxide; Vc = pulmonary capillary blood volume; DmCO = alveolar-capillary membrane diffusing capacity for carbon monoxide.

†Mean (SD) (95% CI). The mean change is follow-up (posttraining) minus baseline.

pulmonary DLCO and DLNO were 42.3 (8.9) and 209 (40) ml·min⁻¹·mm Hg⁻¹, respectively, and were significantly higher compared with predicted values (~115% predicted, $p \leq 0.05$ compared with predicted). One woman (9%) and 9 men (64%) were above the upper limit of normal (95th percentile) for predicted values for DLCO, representing 40% of the 25 subjects. Three women (27%) and 8 men (57%) were above the upper limit of normal for DLNO, representing 44% of the 25 subjects. Thus, nearly half of the swimmers had superior diffusing capacities. The mean DLNO to DLCO ratio was 5.0 (0.3) units, Vc was 94 (17) ml, alveolar volume was 8.3 (1.4) L, and DmCO and DmNO were 212 (49) and 419 (97) ml·min⁻¹·mm Hg⁻¹, respectively. Four weeks of CFB training did not alter any component of pulmonary diffusing capacity (Table 4). The control group, as a whole, also did not have any changes compared with baseline (Table 4). We also compared the number of swimmers in each group who improved by equal to or more than the smallest measurable change of 9 ml·min⁻¹·mm Hg⁻¹ for DLNO and 2.2 ml·min⁻¹·mm Hg⁻¹ for DLCO (23). The control group had 2 and 3 out of 9 swimmers who improved by equal to or more than the smallest measurable change for DLNO and DLCO, respectively; the CFB group had 4 and 5 out of 9 swimmers who improved by equal to or more than the smallest measurable change for DLNO and DLCO, respectively. Thus, there was no significant difference in the proportion of swimmers who improved by at least smallest measurable change for diffusing capacity (Fisher's exact test, $p > 0.32$ for both).

DISCUSSION

The main purpose of this study was to determine to what extent CFB training would mitigate inspiratory muscle

fatigue in a competitive swimming cohort. We demonstrated that there was a 12% decrease in MIP scores after an all-out 200-yard swim race lasting nearly 2 minutes. This drop in MIP is smaller than values of 21, 22, and 29% reported elsewhere for 200 LCM front crawl race-paced efforts (17,27,28). However, critical velocity and its role in respiratory muscle fatigue was studied by Lomax et al. where they found performing to 102% personal record in the 200 m free-style time trial resulted in 22% drop in MIP; however, a performance at 108% only elicited a significant 8% drop in MIP. Our athletes performed to a similar effort (107%) and experienced greater fatigue (12%) racing in SCY format vs. LCM.

After intervention, the experimental group demonstrated no significant drop in MIP scores after the 200-yard time trial. Conversely, the control group experienced the same decrease in MIP scores postexercise as during the baseline session. It is possible that the drop in MIP could represent central fatigue; however unlikely, given the presence of central fatigue would have contributed to a decrease in MIP values at both pre- and posttraining across both groups. Despite a reduction in inspiratory muscle fatigue, neither performance benefits nor pulmonary function was increased. It is possible that a 4-week intervention period was not long enough to influence performance variables, especially during the swimmers' "preseason."

It is unlikely that the adaptations occurred at the pulmonary level given a possible ceiling effect with this subject cohort. Twelve out of 14 male swimmers and 10 out of 11 female swimmers were classified as "excellent" or "superior" in terms of aerobic capacity (milliliters per kilogram per minute) (2). In this study, aerobic capacity was moderately associated with pulmonary diffusing capacity,

measured at rest ($\dot{V}O_2\text{max}$ [milliliters per kilogram per minute] vs. DLCO/body surface area, $r = 0.63$, $p < 0.01$; $\dot{V}O_2\text{max}$ [milliliters per kilogram per minute] vs. DLNO/BSA, $r = 0.53$, $p < 0.01$). This is not uncommon as sedentary unfit populations (11), obese populations (43), and the highly fit (43) show associations between aerobic capacity and pulmonary diffusing capacity, measured at rest. Thus, if aerobic capacity could improve, diffusing capacity could also potentially improve. Aerobic training has shown to improve either DLNO or DLCO (11,14,20).

With respect to possible cross-training adaptations, pulmonary diffusing capacity has been shown to be associated with marathon running performance in those who complete it in approximately 3 hours or less (24,29). The present study also demonstrated strong associations between pulmonary diffusing capacity and 200-yard swim performance, an event that is much shorter in length compared with marathon running. However, the oxygen cost of running was not altered with the CFB intervention in this study despite significantly improving by 6% after a CFB program in a sub-competitive population (23).

Several mechanisms could explain a reduction in respiratory muscle fatigue in the CFB group. One proposed mechanism of action can be derived from a study evaluating the mechanics of breath holding. In trained apnea divers, the actions of the respiratory muscles during extended breath holds can be divided into 2 phases: an *easygoing* phase and a *struggle* phase (9). During a single breath hold that lasts approximately 209 seconds, 55% (115 seconds) was spent during the easygoing phase and congruently 45% (94 seconds) in the struggle phase. During the struggle phase of breath holding, there is a progressive pressure development against the glottis creating higher elastic loading, resulting in increased muscular recruitment of both inspiratory (diaphragm and rib cage muscles) and expiratory muscles (abdominal wall). Cross et al. demonstrate that during the final 40% of the struggle phase (the last ~38 seconds), recruitment of the inspiratory rib cage musculature is preferred over the diaphragm to resist diaphragmatic fatigue (9). During CFB training, each 50-m swim typically lasted approximately 40 seconds. As intervals progressed downward and time spent in rest with unrestricted breathing decreased, the perceived exertion of the CFB group increased. Therefore, we propose that it is possible that the swimmers spent significantly more time in the struggle phase of each breath hold. During training, an increased exposure to the struggle phase may have aided the CFB group to selectively recruit rib cage musculature over the diaphragm to preserve normal diaphragm function, therefore resisting fatigue during the 200-yard time trial.

Larger bouts of high-intensity work (8–10) minutes can typically induce low-frequency fatigue where recovery of the diaphragm takes more than 60 minutes (18). There has been a traditional divide between high- and low-frequency muscle fatigue and a difference between peak power and maximum contraction duration (19). With the rate of fatigue recovery

under 1 minute in the present study and because a maximal static effort is associated with high neuronal frequency, this type of power reduction is typical of high-frequency muscle fatigue ((1), pp. 574).

Many competitive swim programs use 1 or 2 practices per week with multiple maximal efforts, termed “ $\dot{V}O_2$ ” workouts. Repeated exposure to this practice style could have led to high adaptability of the diaphragm and supporting musculature. In fact, recent work has demonstrated that CFB training (twice per week for 5 weeks at 12–20 bouts of 25 m) can improve swimming performance, but the breath holding has to be at functional residual capacity (FRC) or a bit below FRC (40). In the present study, swimmers held their breath at TLC during the swimming workouts to limit the hypoxemia during exercise (39).

PRACTICAL APPLICATIONS

In conclusion, this study demonstrated 3 important findings: First, a 200-yard front crawl time trial is enough to elicit inspiratory muscle fatigue. Second, CFB swimming training appears to prevent high-frequency inspiratory muscle fatigue in competitive college swimmers. However, this did not result in any improvement in swimming performance. Finally, the adaptations incurred by this type of training are suspected to be linked with increased accessory muscle recruitment during breathing.

If the benefits are largely muscular in nature, it is possible that CFB training may be valuable across multiple disciplines where a possible cardiovascular ceiling is present (e.g., running, cycling, rowing). Presently, this study supports the efficacy of using CFB training for NCAA athletes during swim workouts. It is feasible that after a competitive season of using CFB training 3–4 times per week, performance may be increased in comparison against a team not incorporating CFB training.

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